

EBU6408 Sensors and Radio Frequency Identification (RFID)

Lecture 4: Checksum, Coding and Modulation



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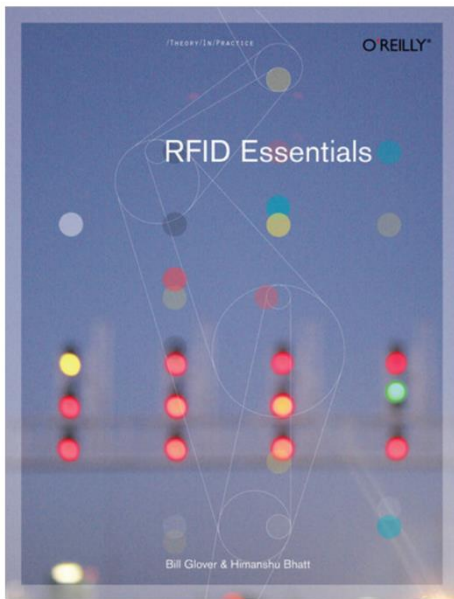
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Textbooks

Bill Glover and Himanshu Bhatt, “RFID Essentials”, O’Reilly Media Inc, 2006.



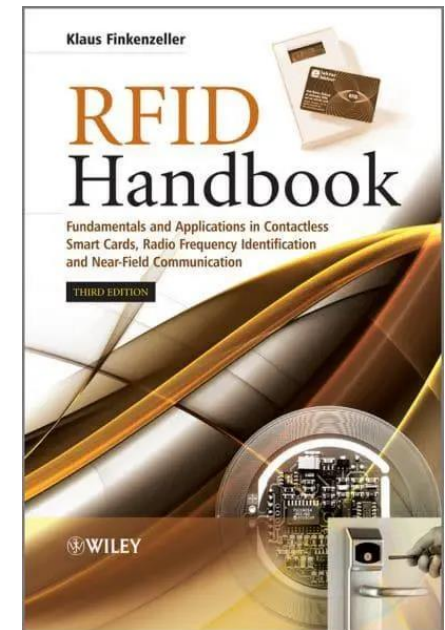
[QMUL Library link](#)

Harvey Lehpamer, “RFID Design Principles”, Artech House, 2012.



[QMUL Library link](#)

Klaus Finkenzeller, “RFID Handbook”, Wiley, Third Edition, 2010.



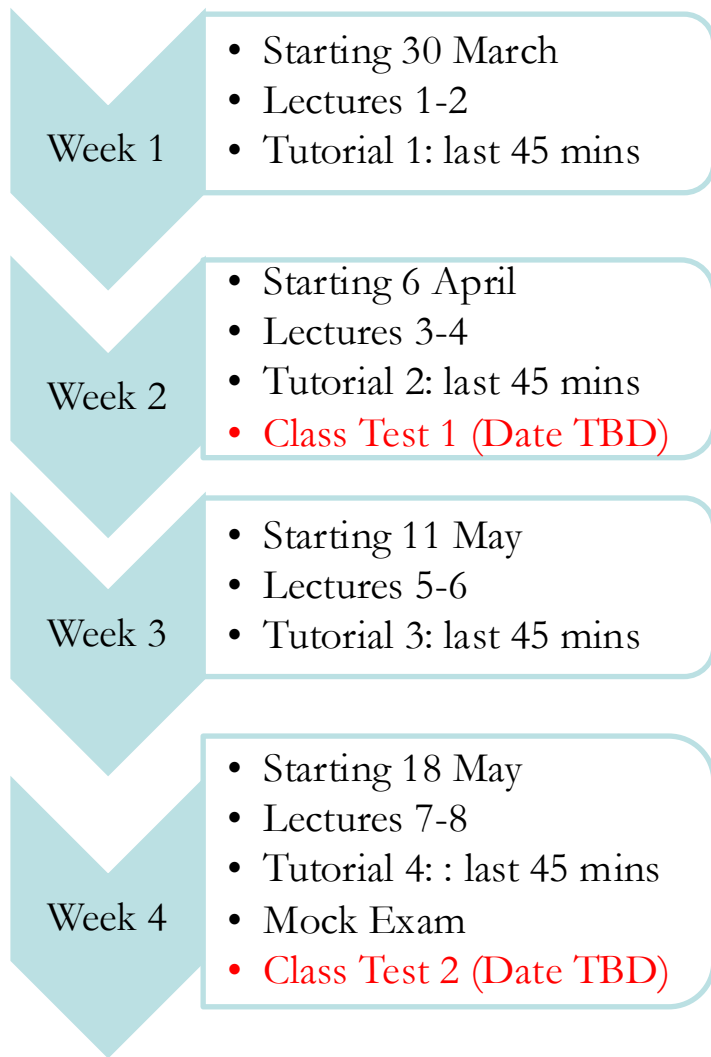
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Evaluation Rules???

- **Two class tests:**
 - **Test 1:** Covering the material delivered in the first half of module **(5%)**
 - **Test 2:** covering the material delivered in the second half of module **(10%)**
 - **Allowed: : Handwriting double-faced A4-sized cheating sheet**
- **Coursework:** Work in groups of 4/5 students.
 - **Lab 1:** Introduction to Arduino **(Bonus Mark)**
 - ✓ Submission of written report on hardware design and programming code.
 - **Lab 2:** RFID projects using Arduino kits **(5%)**
 - ✓ Submission of written report on hardware design and programming code and **5-min video demonstration**
 - **Mini-project:** using practical knowledge from labs and chatbot support **(10%)**
 - ✓ Submission of written report on hardware design and programming code and **10-min video demonstration**
- **Final exam:** covering all teaching weeks **(70%)**
 - **Allowed: Handwriting double-faced A4-sized cheating sheet**
- A couple of QM+ online quizzes on teaching lectures **(Bonus Mark)**



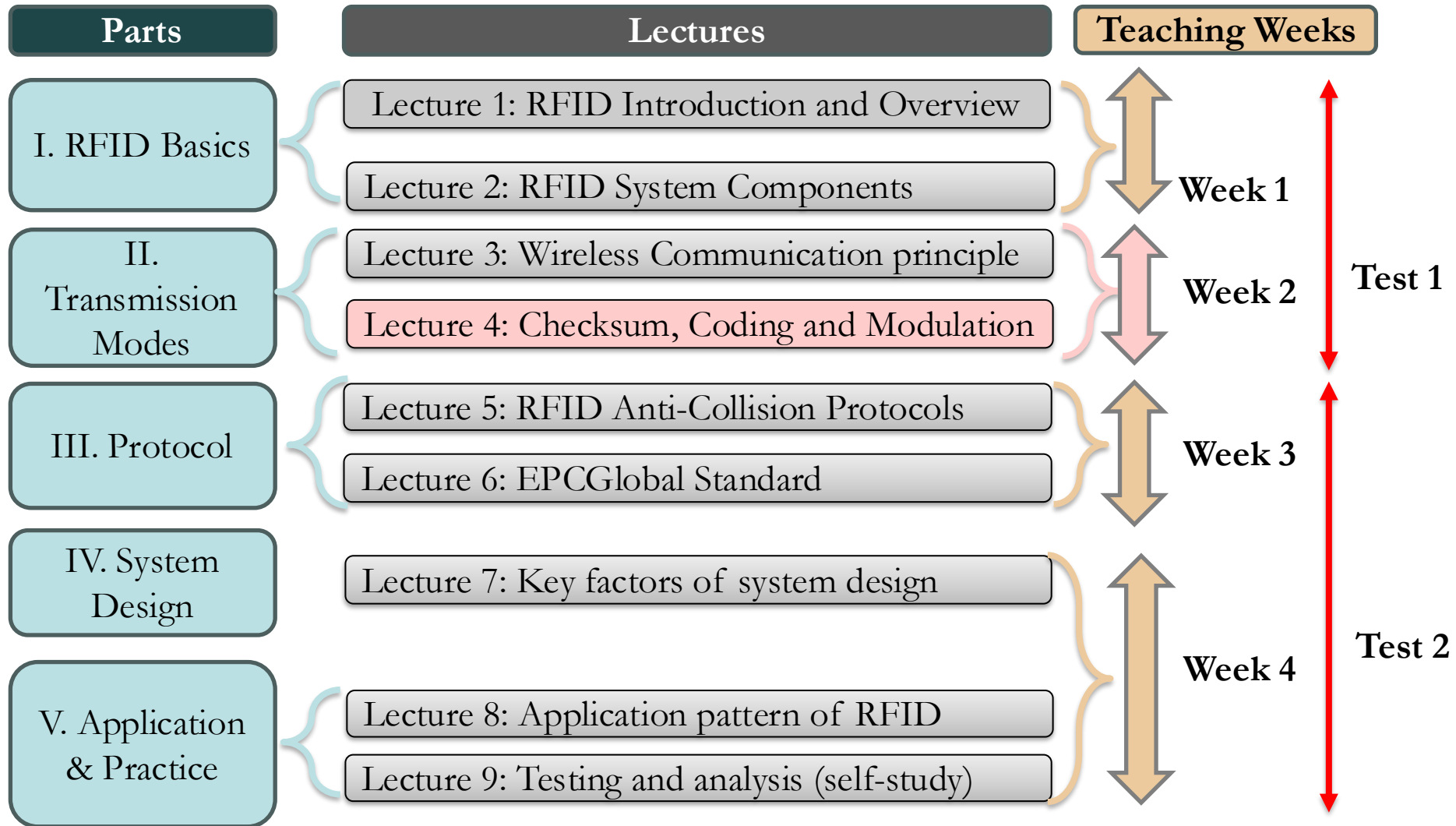
Module Activities



- Labs (Arduino):
 - Two lab sessions will take place in between the teaching weeks.
 - Work in groups (4/5 students) to share the hardware
 - Assignment submission deadlines: usually after lab session.
 - Mini-project due at the end of the term.
- Exact dates will be confirmed later on.

QM+ Quizzes will take place during term time for bonus marks

Module Content



Outline

- Checksum procedures
- Coding and modulation procedures



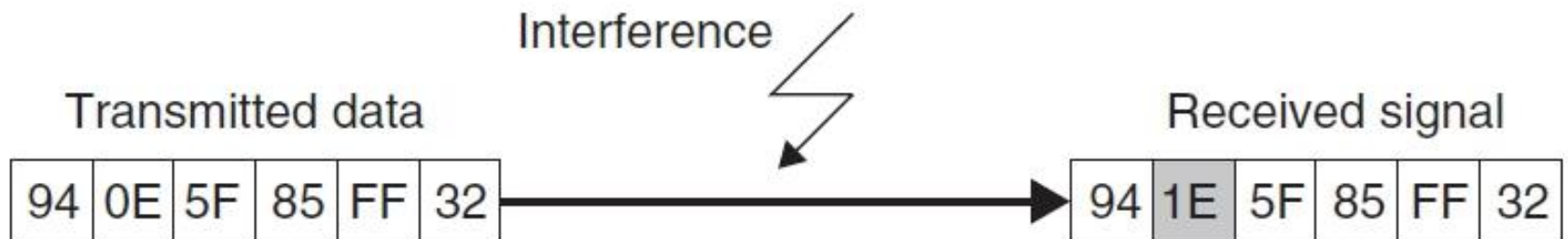
Section 1

CHECKSUM PROCEDURE



The Checksum Procedure

- Checksum
 - Recognize transmission errors
 - Initiate corrective measures: retransmission
- The most common checksum procedures
 - Parity checks
 - LRC
 - CRC



Interference during transmission can lead to errors in the data

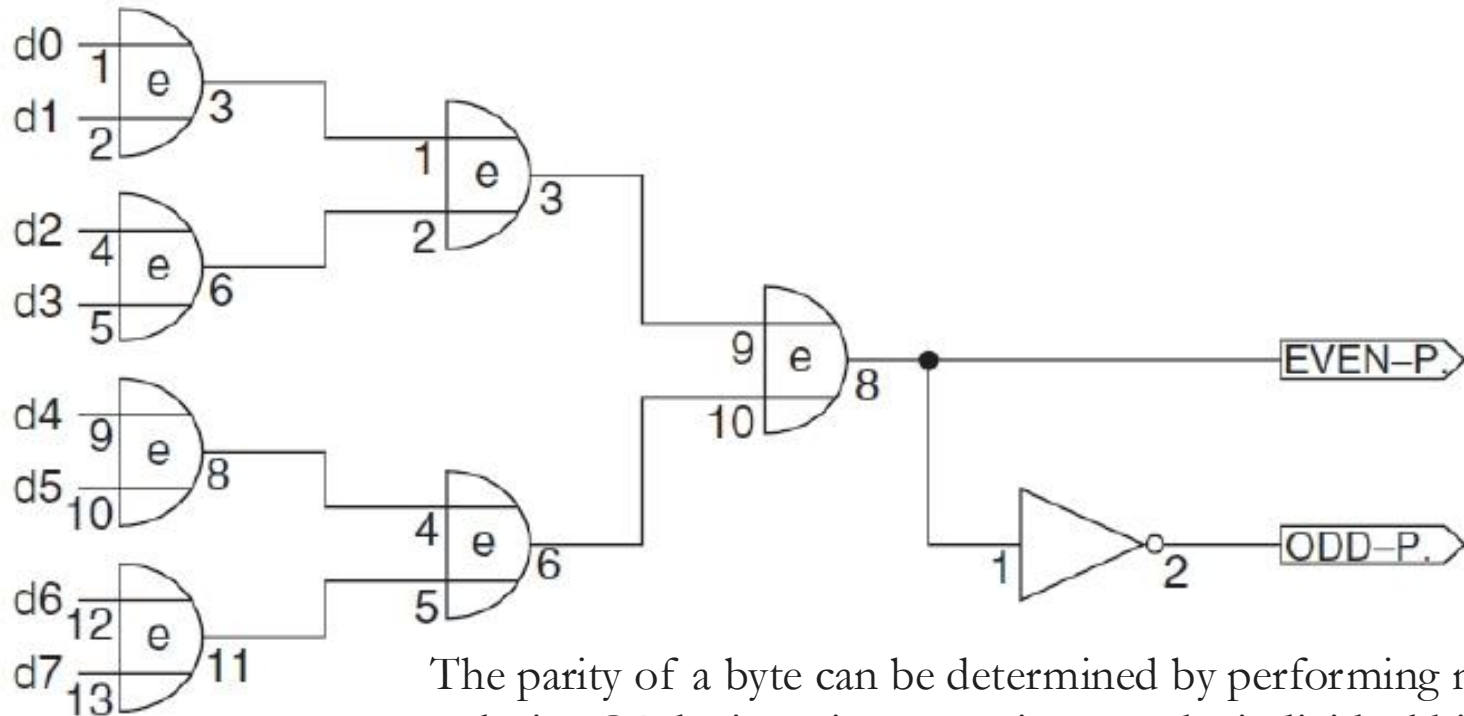
Parity Checking

- A parity bit is incorporated into each byte
- 9 bits are sent for every byte
- Odd parity: an odd number of the 9 bits have the value 1
- Even parity: an even number of the 9 bits have the value 1
- Simple
- Poor error recognition
 - An odd number of inverted bits (1,3,5, ...) : detected
 - An even number of inverted bits (2,4,6,...) : the errors cancel each other out, appear to be correct.



Parity Checking (Cont'D)

- Odd parity: E5h, 1110 0101 $p=0$.
- A parity generator for **even parity**: the **XOR** logic gating of all the data bits in a byte
- **Odd parity**: invert the output



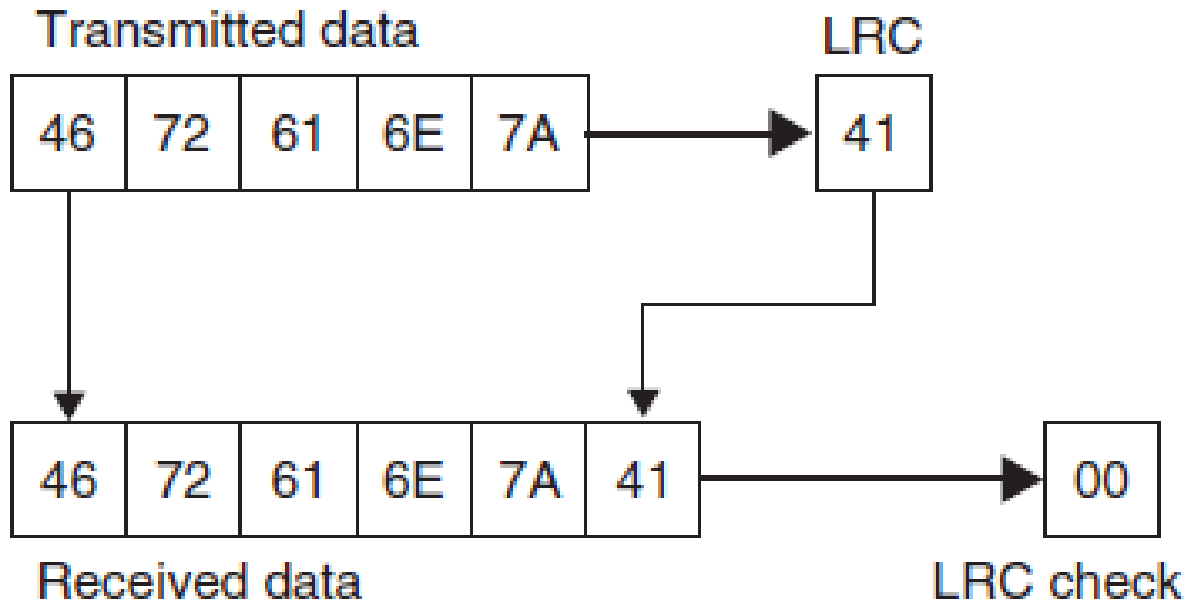
The parity of a byte can be determined by performing multiple exclusive OR logic gating operations on the individual bits

LRC Procedure

- LRC: Longitudinal redundancy check
- The XOR checksum is generated by the recursive XOR gating of all the data bytes in a data block.
- Byte 1 is XOR gated with byte 2, the outcome of this gating is XOR gated with byte 3, and so on.
- If the LRC value is appended to a data block and transmitted with it, then a simple check for transmission errors can be performed in the receiver by generating an LRC from the data block + LRC byte.
- The result of this operation must **always be zero**; any other result indicates that transmission errors have occurred.

LRC Procedure (Cont'D)

- **Can be calculated very simply and quickly**
- **Not very reliable** because it is possible for multiple errors to cancel each other out, and the check cannot detect whether bytes have been transposed within a data block
- **Primarily used for the rapid checking of very small data blocks (e.g. 32 byte)**



46	0100	0110
72	0111	0010
	0011	0100
61	0110	0001
	0101	0101
6E	0110	1110
	0011	1011
7A	0111	1010
41	0100	0001

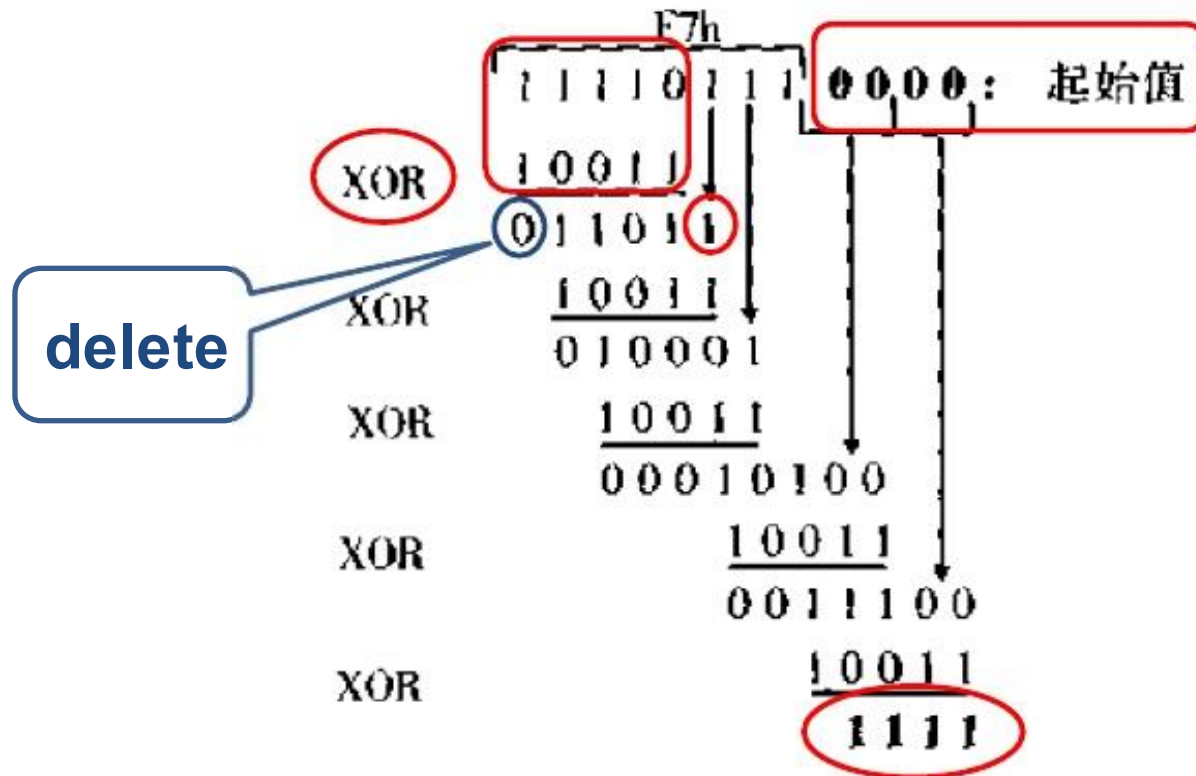
CRC Procedure

- **CRC (cyclic redundancy check) procedure** : originally used in disk drives
- Excellently suited for error recognition
- Highly reliable method of **recognizing** errors
- **Cannot correct** errors
- A CRC checksum is calculated by the division of a polynomial using a so-called *generator polynomial*.
- The CRC value is the remainder obtained from this division.



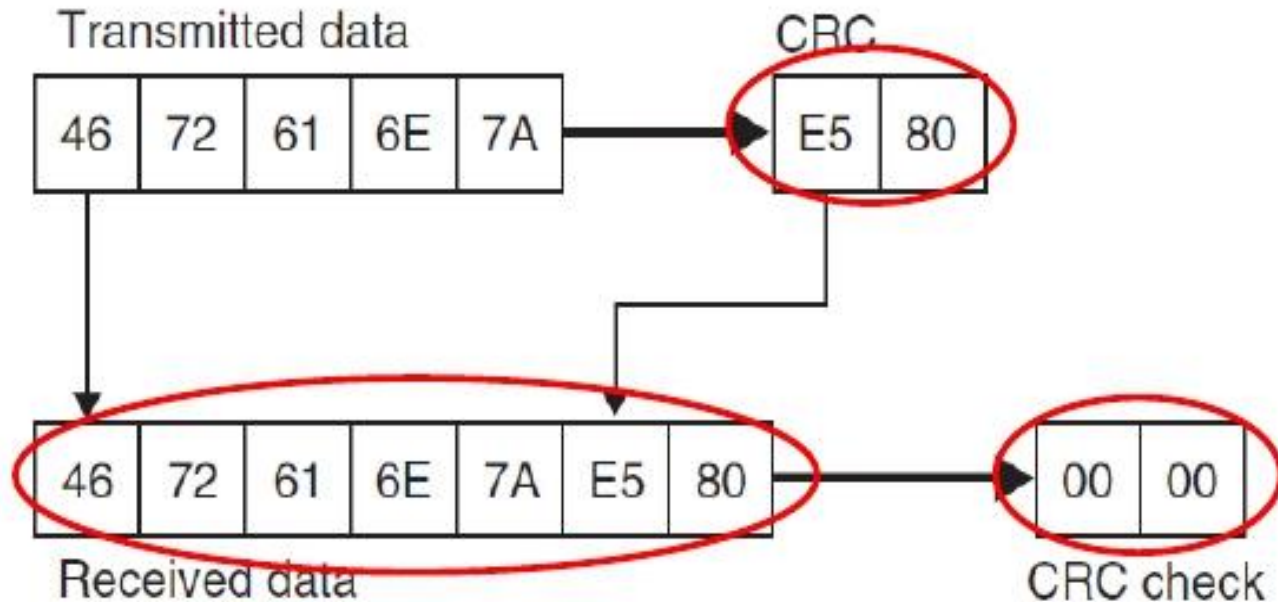
CRC Procedure (Cont'D)

- **CRC (cyclic redundancy check) procedure** : originally used in disk drives
 - The first byte of the data block is F7h=11110111
 - The generator polynomial is $x^4 + x + 1 = 10011$
 - The starting value : 0000B



CRC Procedure (Cont'D)

- Transmitter: calculate the CRC value, append to data block
- Receiver: calculate the CRC value of the received data, including the appended CRC byte
- The result is always zero, unless transmission errors
- Both CRC calculations start from the same initial value



CRC Procedure (Cont'D)

- Advantage: reliability of error recognition
- A16-bit CRC : checking the data integrity of data blocks up to 4Kbytes in length
- RFID systems : data blocks, shorter than 4Kbytes, 12- and 8-bit CRCs

CRC-8 generator polynomial

CRC-16 disk controller generator polynomial

CRC-16 CCITT generator polynomial

$$x^8 + x^4 + x^3 + x^2 + 1$$

$$x^{16} + x^{15} + x^2 + 1$$

$$x^{16} + x^{12} + x^5 + 1$$

Examples of different generator polynomials

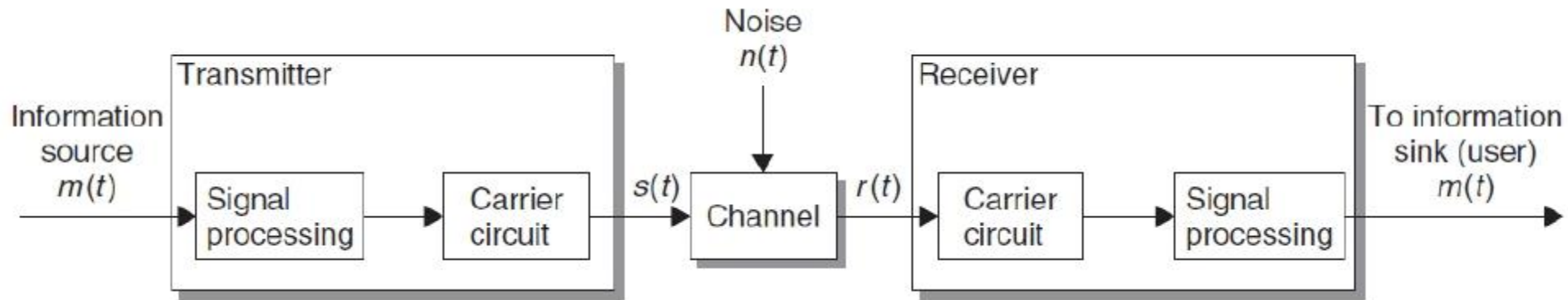
Section 2

CODING AND MODULATION



Coding and Modulation

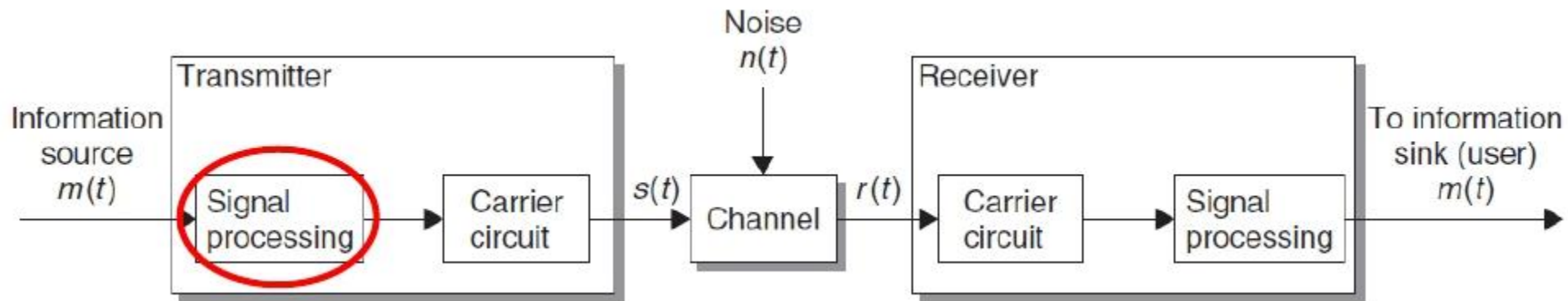
- From the reader to the transponder
 - Signal coding the modulator in the reader (transmitter)
 - The transmission medium (channel)
 - The demodulator and signal decoding in the transponder (receiver)



Signal and data flow in a digital communication system

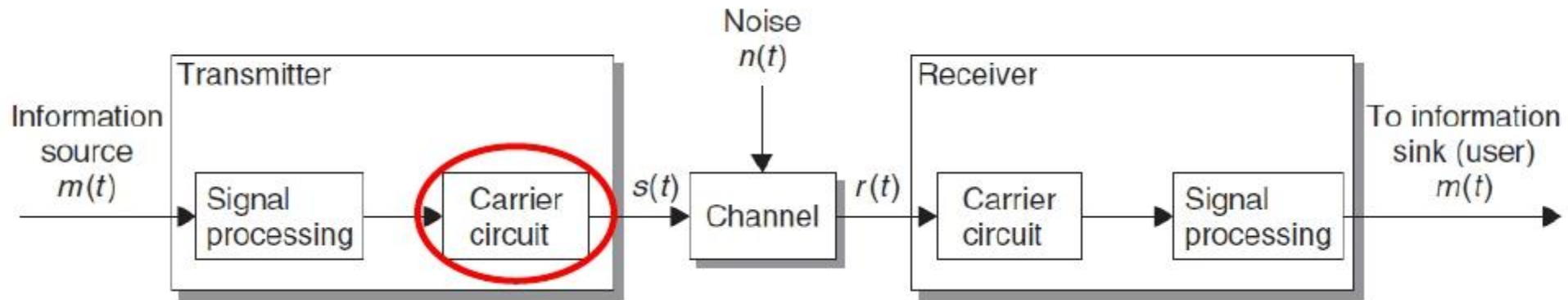
Signal Coding

- Matches the message optimally to the characteristics of the transmission channel
- Protection against interference or collision
- Protection against intentional modification
- Coding in the baseband



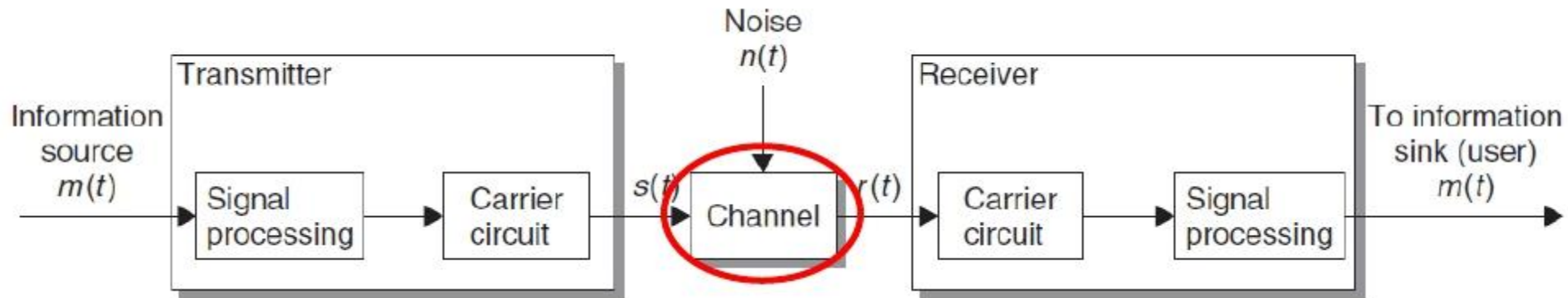
Modulation

- Altering the signal parameters of a high frequency carrier
- Amplitude
- Frequency
- Phase



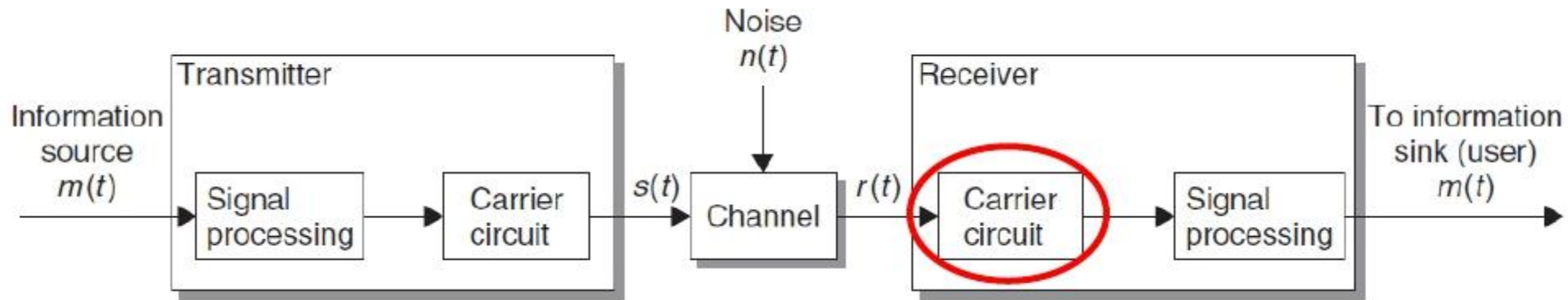
Communication Medium

- Transmits the message over a predetermined distance
- Magnetic fields (inductive coupling)
- Electromagnetic waves (electromagnetic backscatter coupling)



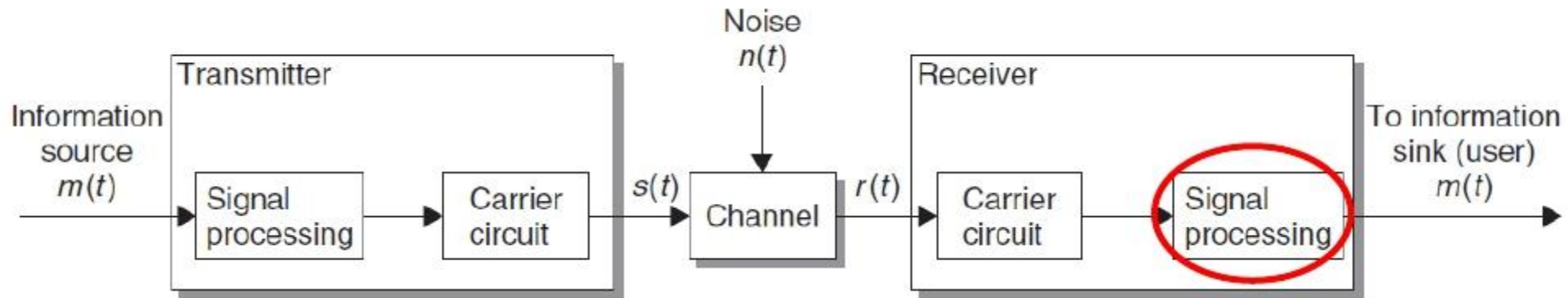
Demodulation

- Reclaim the signal in the baseband
- Contain both a *modulator* and a *demodulator*
- *Modem* (**m**odulator – **d**emodulator)



Decoding

- Reconstruct the original message from the baseband-coded received signal
- Recognize any transmission errors and flag them



Coding in the Baseband

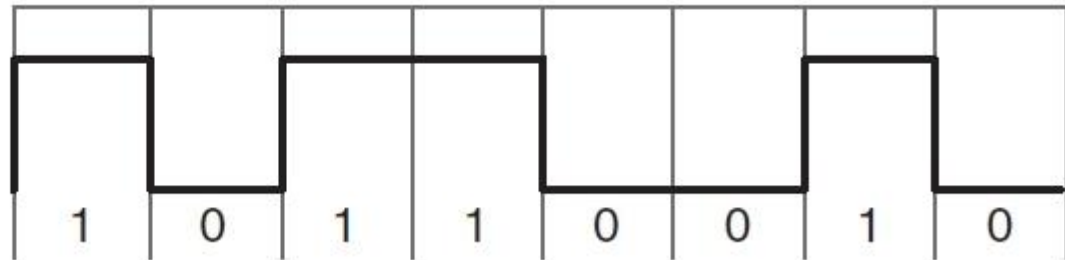
- RFID systems normally use one of the following coding procedures:
 - NRZ coding (Not Return to Zero)
 - Manchester coding
 - Unipolar RZ coding
 - DBP (differential bi-phase)
 - Miller coding
 - Differential coding
 - PP (pulse pause) coding



Coding in the Baseband

- NRZ coding (Not Return to Zero)
 - Binary 1 : a 'high' signal
 - Binary 0 : a 'low' signal

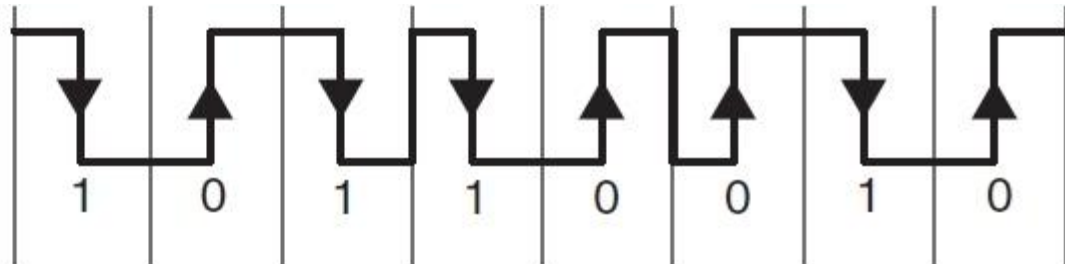
NRZ coding:



Coding in the Baseband

- Manchester Coding
 - binary 1: a negative transition
 - binary 0: a positive transition
 - used for data transmission from the transponder to the reader, based upon load modulation using a subcarrier

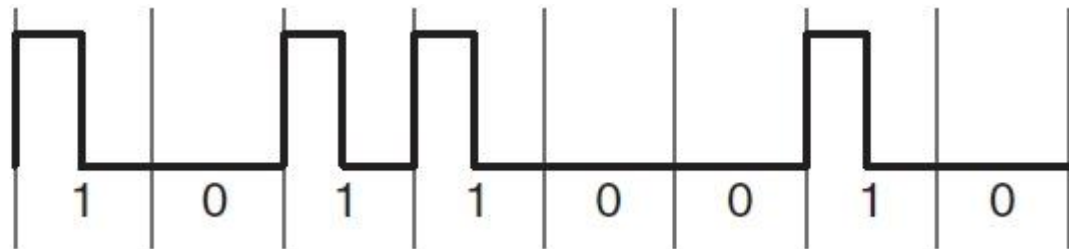
Manchester coding:
(bi-phase)



Coding in the Baseband

- Unipolar RZ Coding
 - Binary 1 : a 'high' signal during the first half-bit period
 - Binary 0 : a 'low' signal lasting for the entire duration of the bit

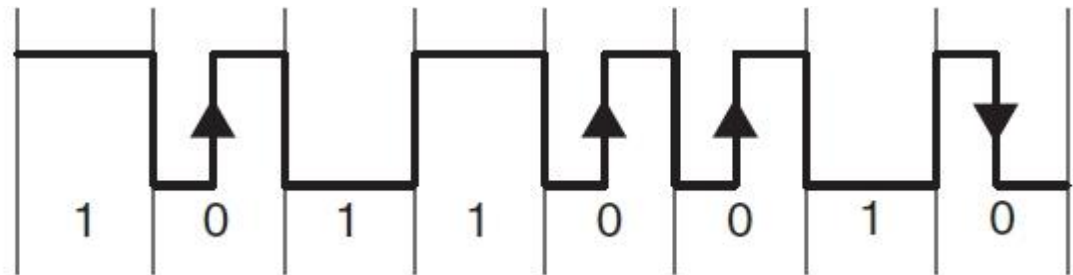
Unipolar RZ coding:



Coding in the Baseband

- DBP (differential bi-phase)
 - Binary 0: a transition of either type in the half-bit period
 - Binary 1: lack of a transition
 - The level is inverted at the start of every bit period
 - The bit pulse can be more easily reconstructed in the receiver

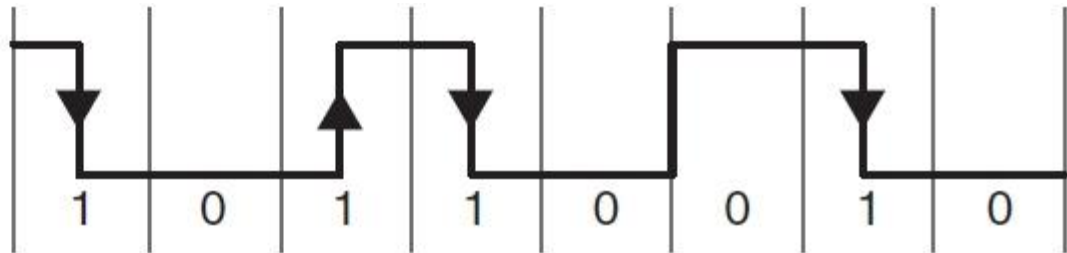
DBP



Coding in the Baseband

- Miller coding
 - Binary 1 : a transition of either type in the half-bit period
 - Binary 0 : continuance of the 1 level over the next bit period
 - A sequence of zeros : transition at the start of a bit period
 - The bit pulse can be more easily reconstructed in the receiver

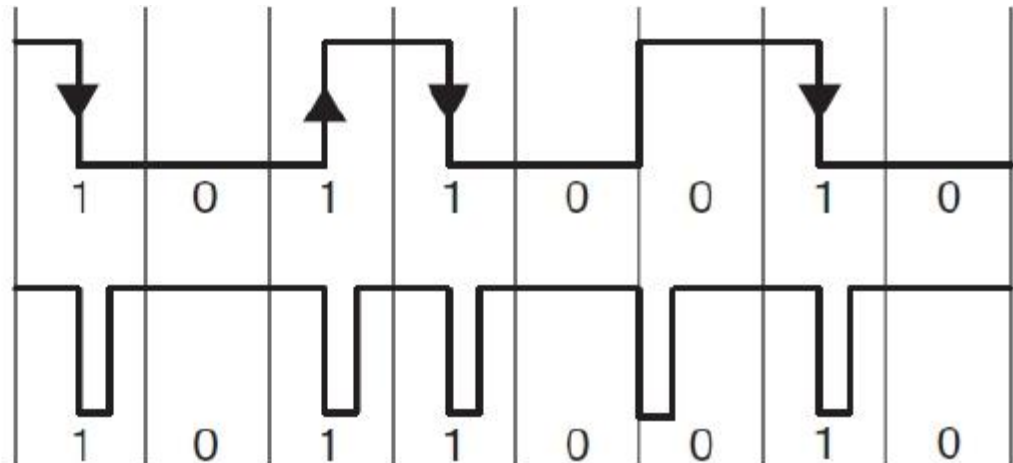
Miller coding:



Coding in the Baseband

- Modified Miller coding
 - each transition : a negative pulse
 - highly suitable for use in **inductively coupled RFID** systems : data transfer from reader to transponder
 - $t_{pulse} \ll T_{bit}$, ensure a continuous power supply to the transponder during data transfer

Miller coding:

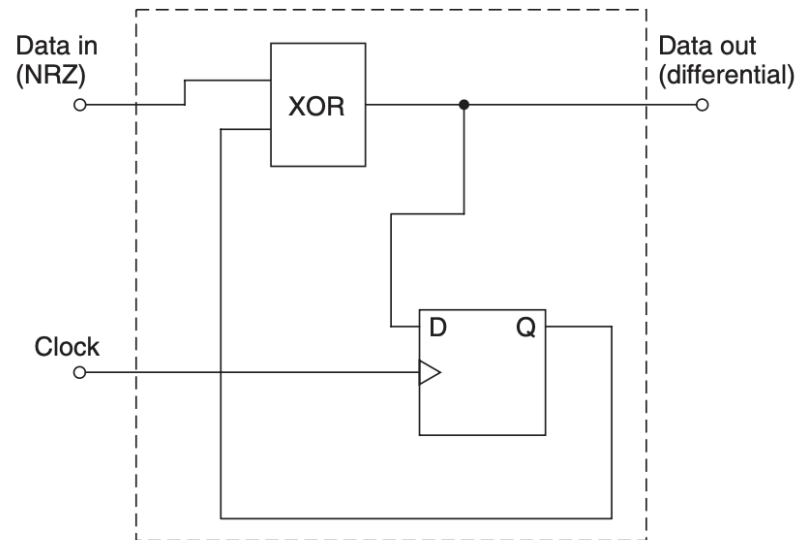
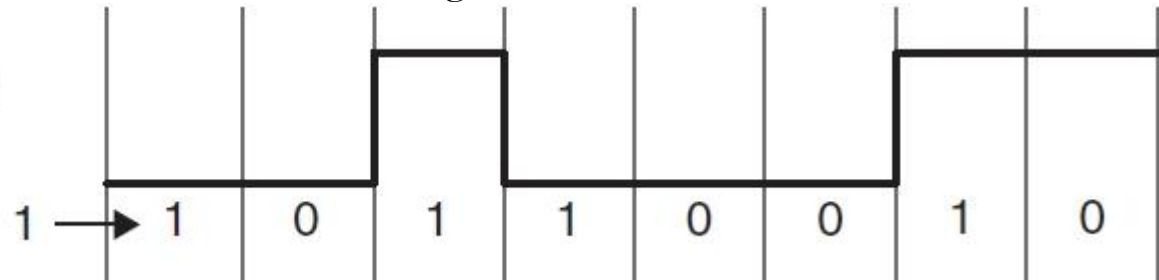


Modified Miller coding:

Coding in the Baseband

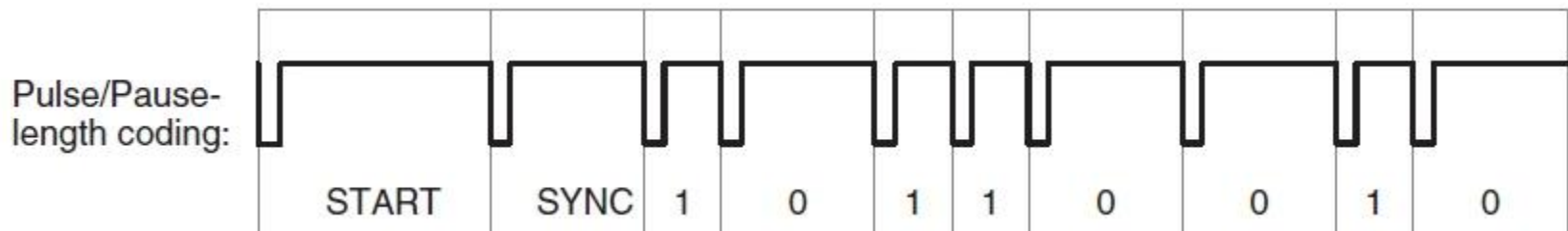
- Differential coding
 - binary 1 : a change (toggle) in the signal level
 - binary 0 : the signal level remains unchanged

Differential coding:



Coding in the Baseband

- Pulse-pause coding (PPC)
 - binary 1 : a pause of duration t before the next pulse
 - binary 0 : a pause of duration $2t$ before the next pulse
- popular in **inductively coupled RFID systems**: data transfer from the reader to the transponder



Coding in the Baseband

- Select a suitable signal coding system
- The most important consideration
 - The signal spectrum after modulation
 - Susceptibility to transmission errors
- In the case of passive transponders,
 - Power supply must not be interrupted by an inappropriate combination of signal coding and modulation procedures.



Digital Modulation Procedures

- Modulation
 - Influencing signal parameters – power, frequency, phase position – of an electromagnetic wave by messages (data)
- Carrier: an unmodulated electromagnetic wave
- Demodulation
 - Reconstruct the message
 - Analyzing the characteristics of an electromagnetic wave
 - Measuring the change in reception power, frequency or phase position of the wave



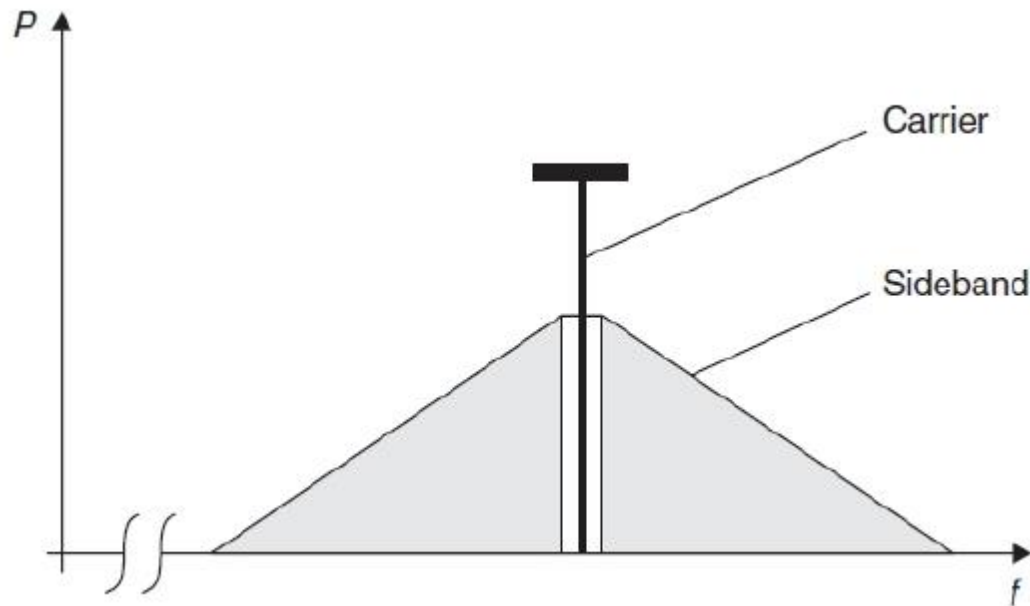
Digital Modulation Procedures

- Analogue modulation
 - Amplitude modulation
 - Frequency modulation
 - Phase modulation
- Digital modulation: used in RFID systems
 - ASK (amplitude shift keying)
 - FSK (frequency shift keying)
 - PSK (phase shift keying)



Digital Modulation Procedures

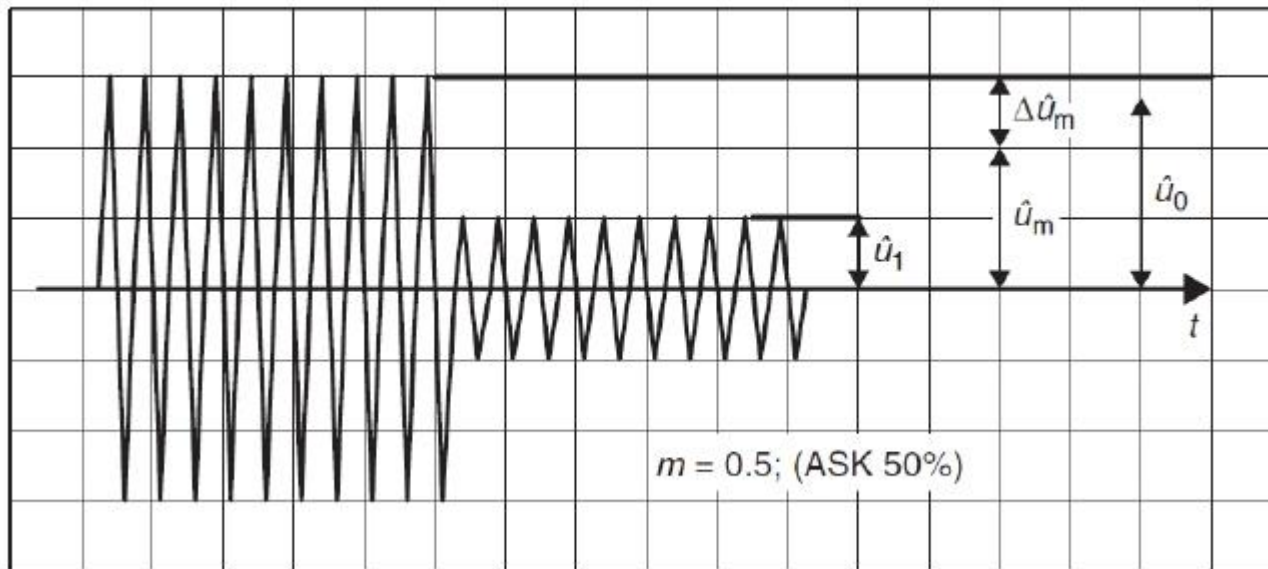
- Symmetric modulation products (sidebands) are generated around the carrier.
- The spectrum and amplitude of the sidebands are influenced by
 - The spectrum of the code signal in the baseband
 - The modulation procedure



Amplitude Shift Keying (ASK)

- Amplitude of a carrier oscillation: switched between two states u_0 and u_1 , by a binary code signal.

- The duty factor m :
$$m = \frac{\Delta u_m}{\hat{u}_m} = \frac{u_0 - u_m}{\hat{u}_m} = \frac{\hat{u}_0 - \hat{u}_1}{\hat{u}_0 + \hat{u}_1}$$

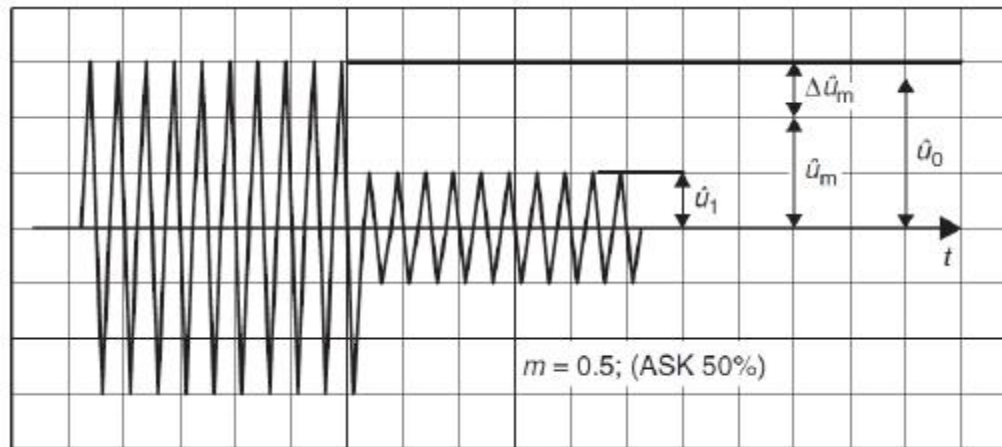


Amplitude Shift Keying (ASK) (Cont'D)

- The duty factor: the percentage carrier reduction m' during keying

$$m' = 1 - \frac{\hat{u}_1}{\hat{u}_0}$$

- In the case of duty factors $< 15\%$ and duty factors $> 85\%$, the differences between the two calculation methods can be disregarded.

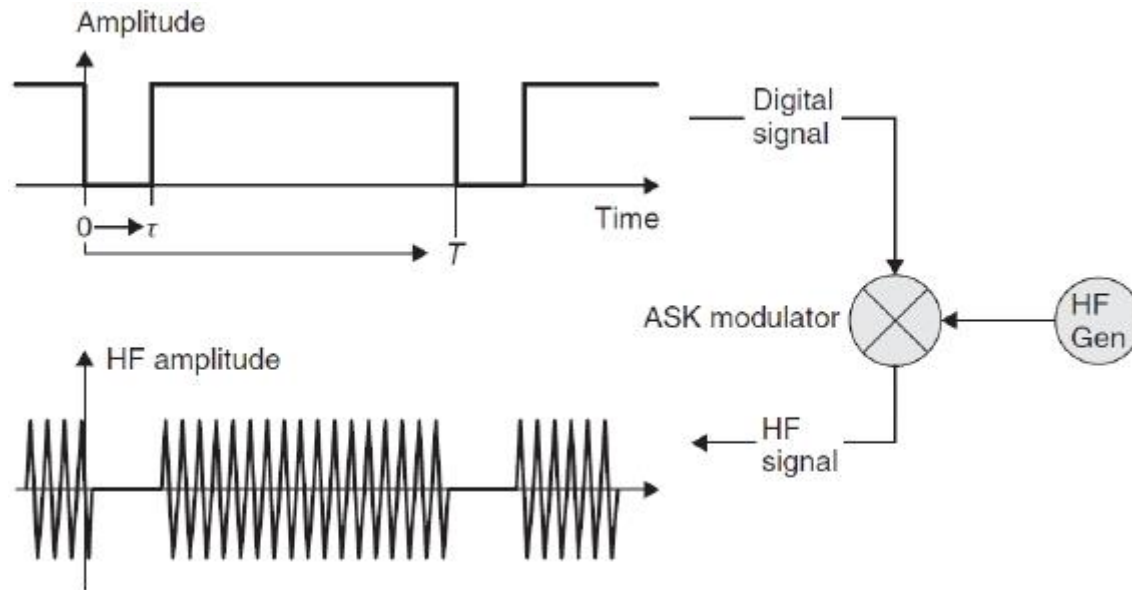


Amplitude Shift Keying (ASK) (Cont'D)

- ASK modulation: multiply the code signal by the carrier oscillation

$$U_{ASK}(t) = (m \cdot u_{code}(t) + 1 - m) \cdot u_{HF}(t)$$

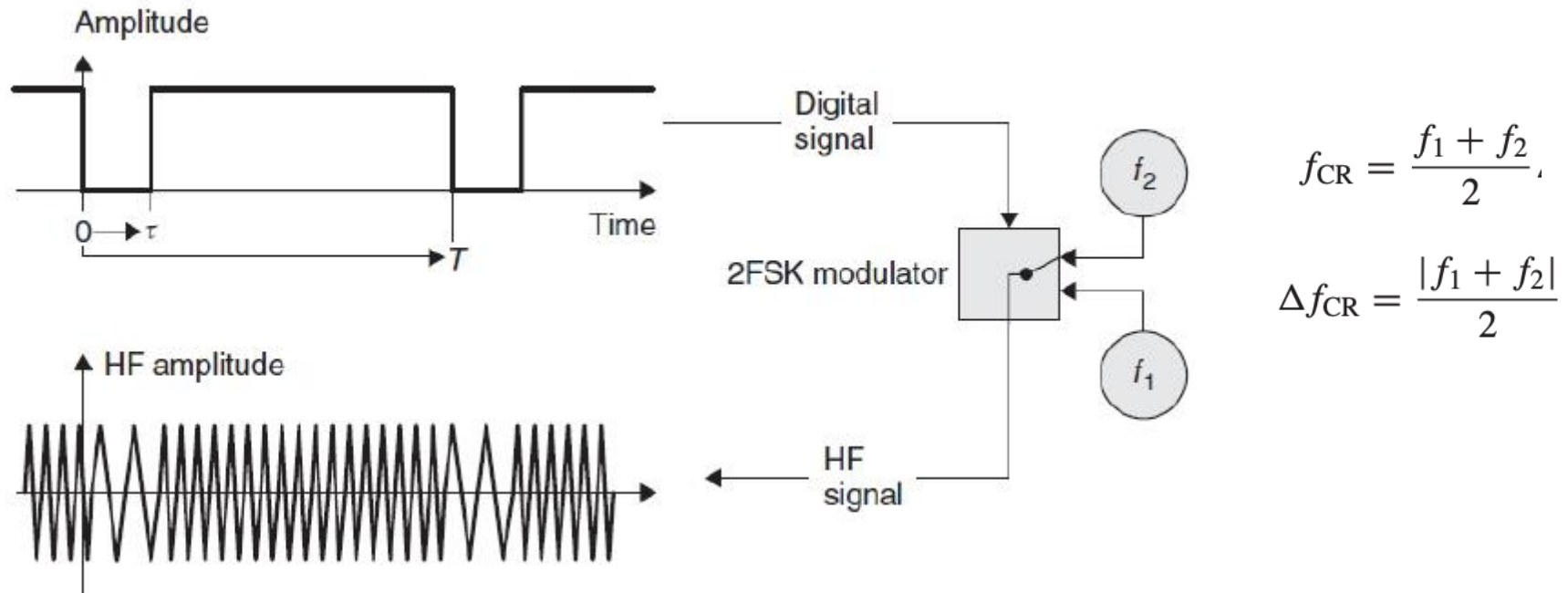
- The spectrum of ASK signals: convolution of the code signal spectrum with the carrier frequency



$m=1$: on-off keying

Frequency Shift Keying (FSK)

- 2FSK, the frequency of a carrier oscillation: switched between two frequencies f_1 and f_2 , by a binary code signal.

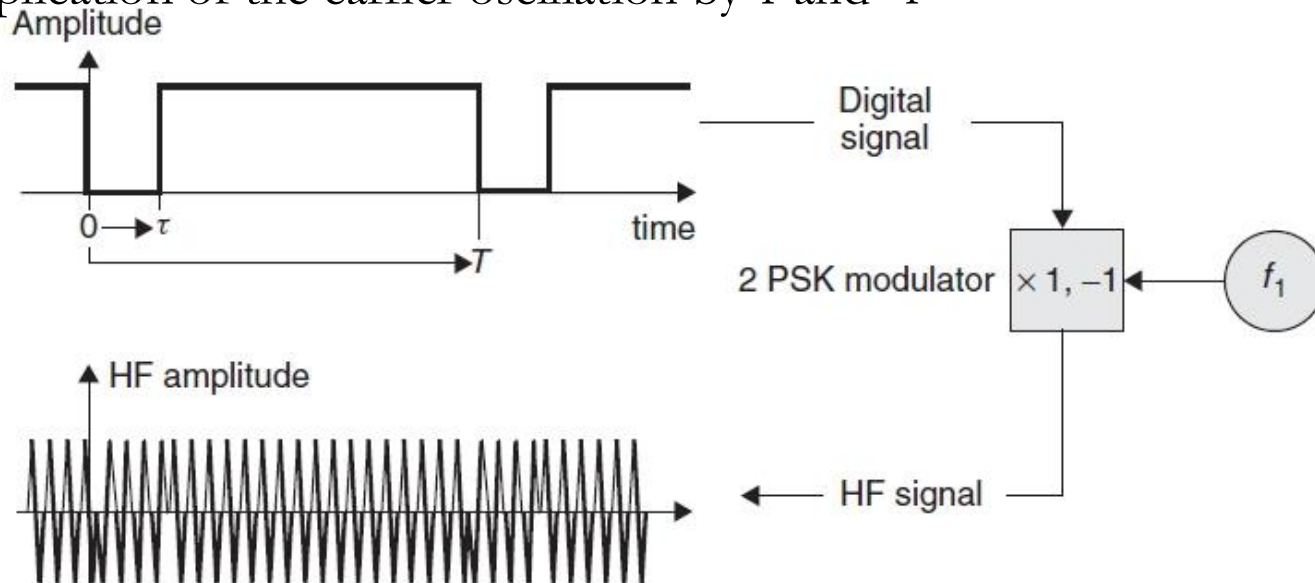


The baseband coding used in RFID systems produces an asymmetric frequency shift keying:

$$\tau \neq \frac{T}{2}$$

Phase Shift Keying (PSK)

- 2PSK: switched between the phase states 0° and 180° corresponds with the multiplication of the carrier oscillation by 1 and -1



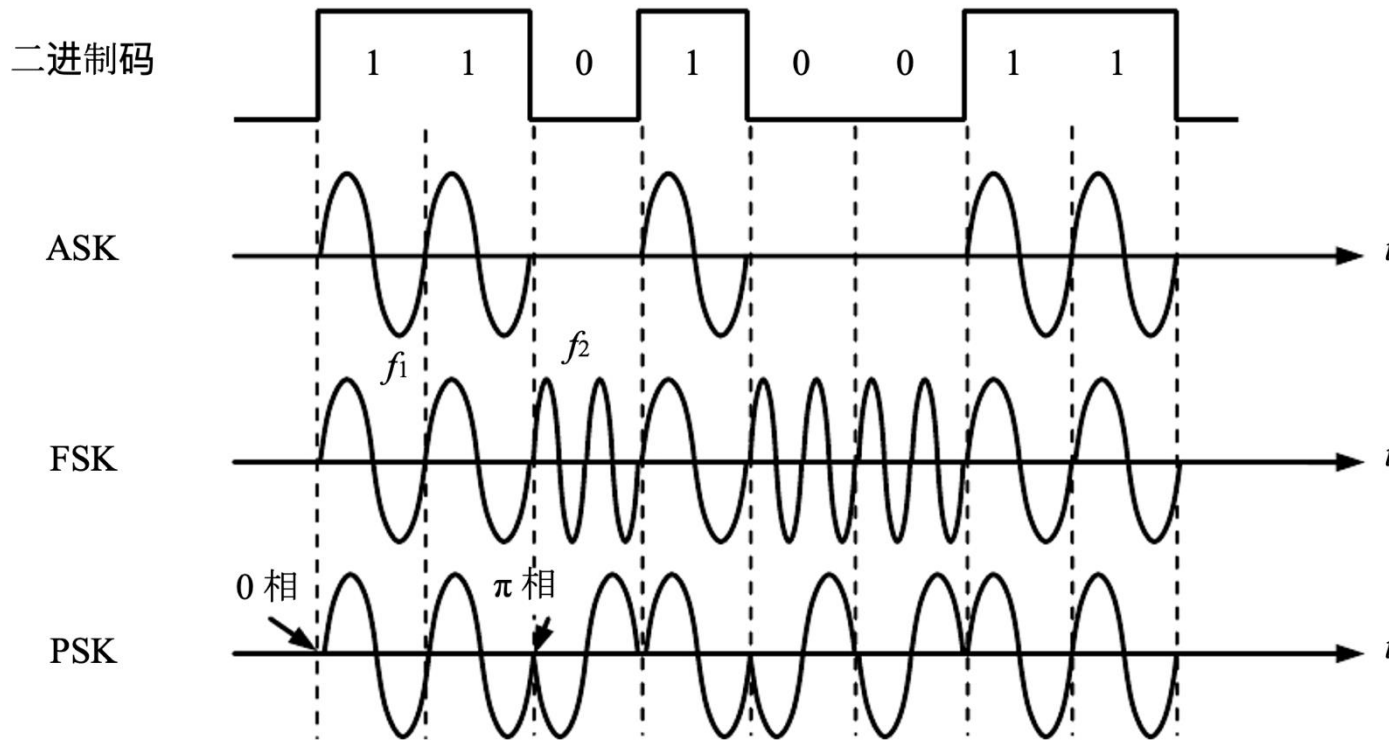
The power spectrum of a 2 PSK can be calculated as follows for a mark-space ratio τ/T of 50%

$$P(f) = \left(\frac{P \cdot T_s}{2} \right) \cdot [\sin^2 \pi(f - f_0)T_s + \sin^2 \pi(f + f_0)T_s]$$

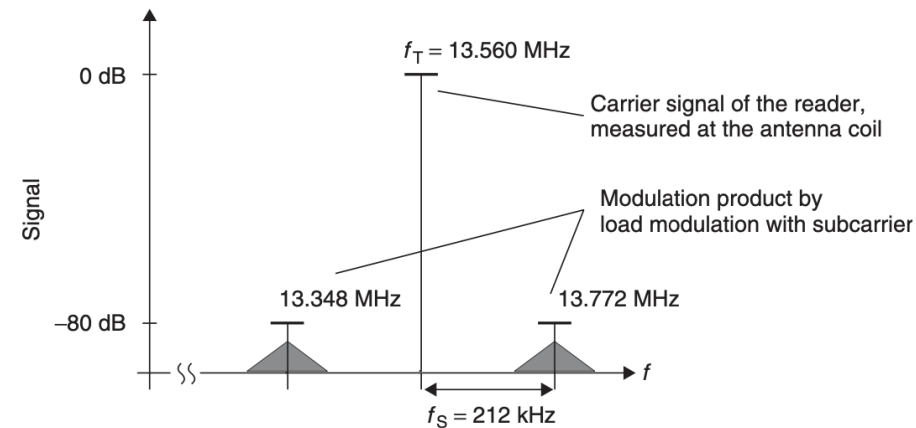
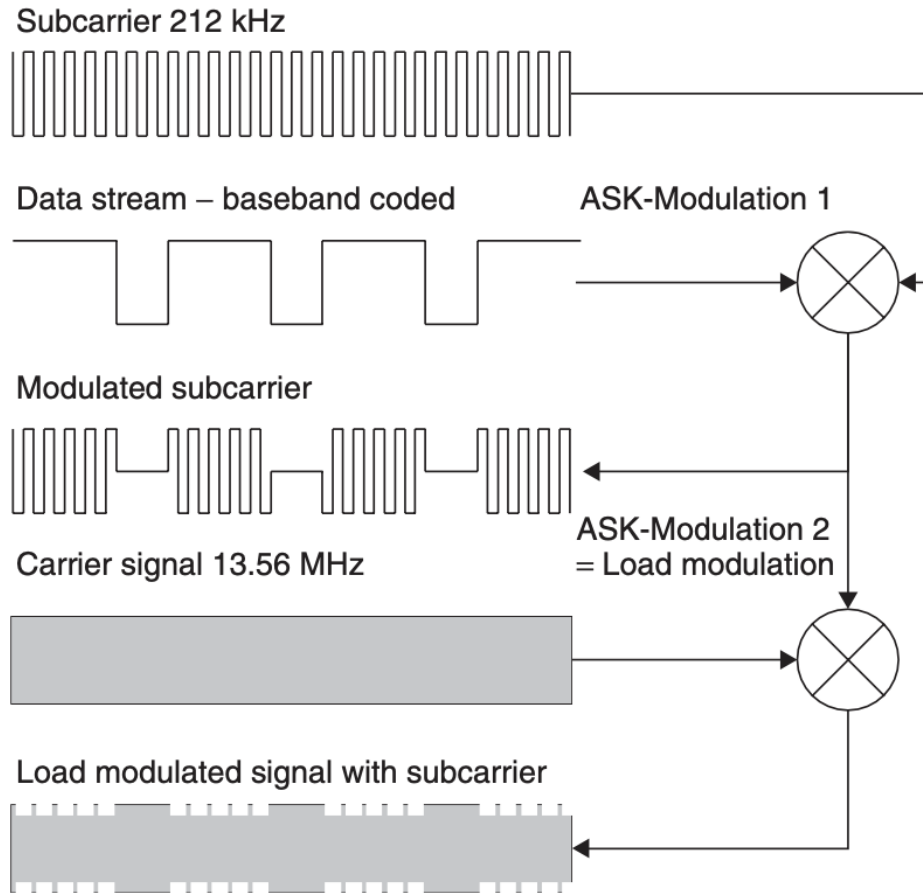
where P is transmitter power, T_s is bit duration ($= \tau$), f_0 is centre frequency, and $\text{sinc}(x) = (\sin(x)/x)$.

Digital Modulation Procedures

- 2PSK: switched between the phase states 0° and 180° corresponds with the multiplication of the carrier oscillation by 1 and -1



Modulation Procedures with Subcarrier



Thank You



Dr Fatma Benkhelifa